

Financial Copilot Platform

Capabilities & Testing Guide — May 2026

Repos: financial-copilot-api · financial-copilot-ai-orchestrator · financial-copilot-web

What exists today

Layer	Implemented	Not in UI / not built
Web	Dashboard, demo seed, streaming	Watches, account forms, auth, extra nav pages
API	Dashboard, accounts, holdings, seed, CSV imports	Feed, CSV exports, reports, bond optimiser, auth
Orchestrator	Health, chat, stream, session management	Dynamic Phase C summarization

Recommended test order

0. Stack up (ports 8080, 8000, 3000) → 1. Health → 2. Demo seed → 3. Dashboard UI → 4. AI basic → 5. AI memory/streaming → 6. Agent routing → 7. Accounts/holdings (API) → 8. CSV imports (API)

0. Prerequisites

Service	URL	Notes
API	http://localhost:8080	Spring Boot + PostgreSQL
Orchestrator	http://localhost:8000	GOOGLE_API_KEY in .env
Web	http://localhost:3000	NEXT_PUBLIC_AI_URL → :8000

Pass when: All three URLs respond without connection errors.

1. Infrastructure & connectivity

#	Capability	How to test	Pass criteria
1.1	API liveness	GET /actuator/health	status "UP"
1.2	Orchestrator	GET /health	status "ok"
1.3	API from orchestrator	GET /health/dependencies	status "ok"
1.4	Web loads	Open :3000	Page renders
1.5	Agents list	GET /api/v1/agents	Lists agents

PowerShell smoke:

```
Invoke-RestMethod http://localhost:8080/actuator/health
Invoke-RestMethod http://localhost:8000/health
Invoke-RestMethod http://localhost:8000/health/dependencies
Invoke-RestMethod http://localhost:8000/api/v1/agents
```

2. Demo data (required first)

#	Capability	How to test	Pass criteria
2.1	Seed demo	UI: "Seed Demo Data"	Metrics + holdings populate
2.2	Reseed	POST /api/v1/demo/seed?reset=true	No 412 on dashboard

Without seed: dashboard and AI fail (412 — demo user not seeded).

3. Dashboard (web UI)

#	Capability	How to test	Pass criteria
3.1	KPI cards	After seed	Net worth, assets, liabilities, health /100
3.2	Region chart	Left pie	Segments + legend %
3.3	Asset class chart	Right pie	Segments + legend %
3.4	Holdings table	Scroll table	Symbols, qty, ZAR values
3.5	Errors	Stop API, refresh	Error banner + retry
3.6	API parity	GET /api/v1/dashboard	Matches UI numbers

Placeholder nav: Portfolio, Retirement, Bonds are disabled — no pages.

4. AI chat — basic (web UI)

#	Capability	How to test	Pass criteria
4.1	Send message	Ask net worth (one sentence)	User + assistant bubbles
4.2	Streaming	Watch reply	Token-by-token growth
4.3	Real data	Compare to dashboard	Figures align
4.4	Brevity	Long question	2–4 short paragraphs
4.5	Agent label	After reply	e.g. portfolio_analysis
4.6	Disclaimer	Panel footer	Educational only text
4.7	Suggestions	Empty-state chips	Sends prefilled question

5. AI chat — session & streaming

#	Capability	How to test	Pass criteria
5.1	Multi-turn	Net worth → "shorter"	Context-aware shorter reply
5.2	Refresh	F5 mid-chat	Thread restored
5.3	New chat	New chat button	Empty thread, new session
5.4	Session API	PowerShell script below	History ≥ 2 msgs
5.5	Non-stream API	POST /api/v1/chat	reply + active_agent
5.6	Stream API	POST /api/v1/chat/stream	token + done events

Session test (PowerShell):

```
$sid = [guid]::NewGuid().ToString()
Invoke-RestMethod http://localhost:8000/api/v1/chat -Method POST ...
Invoke-RestMethod "http://localhost:8000/api/v1/chat/session/$sid"
Second POST with message "shorter" and same session_id
```

Note: Server memory is in-process (MemorySaver). Restarting orchestrator clears server sessions.

6. AI agent routing

Keyword routing; portfolio context loaded from API for specialist routes.

Agent	Test prompt	Pass criteria
portfolio_analysis	How is my allocation?	Uses holdings/allocation
tax	Explain CGT on my gains	Agent label tax
bond_optimisation	Pay extra on home loan?	Uses liabilities
risk_analysis	Concentration risk?	Risk bullets
recommendation	What next for TFSA?	2–3 suggestions
scenario	What if rates rise 2%?	What-if answer
report_generation	Wealth summary report	Brief snapshot
general	Hello	Conversational

7. Accounts & holdings (API only)

#	Capability	Endpoint	Pass
7.1	List accounts	GET /api/v1/accounts	Demo accounts array
7.2	Create account	POST /api/v1/accounts	201
7.3	List holdings	GET .../accounts/{id}/holdings	Holdings list
7.4	Add holding	POST .../holdings	201
7.5	Dashboard sync	Refresh dashboard / AI	New data visible

Swagger: <http://localhost:8080/swagger-ui.html>

8. CSV import pipeline (API only)

See [financial-copilot-api/docs/IMPORTS.md](#) for full workflow.

#	Step	Endpoint / action	Pass
8.1	Source types	GET /imports/source-types	Lists source types
8.2	Preview	POST /imports/preview	Column map suggested
8.3	Upload	POST /imports/upload	Job REVIEW
8.4	Records	GET .../records	VALID/ERROR rows
8.5	Fix	PATCH + revalidate	Errors cleared
8.6	Commit	POST .../commit	Data persisted
8.7	Bond CSV	BOND_STATEMENT sample	Balance updated
8.8	Profiles	mapping-profiles GET/POST	Save/reuse maps
8.9	Cancel	POST .../cancel	Job cancelled

Samples: [backend/portfolio-service/src/main/resources/samples/](#)

Not implemented yet (skip for current release)

- Login / auth / multi-user
- Tax engine REST API (AI discusses tax in prose only)
- Monte Carlo / scenarios API
- Report PDF generation
- Bond optimiser API
- Phase C — long-thread summarization
- Web UI for imports, accounts, settings

Gate checklist before Phase C or deploy

- All services healthy; /health/dependencies ok
- Demo seeded
- Dashboard metrics, charts, holdings correct
- Streaming chat + follow-up "shorter" + refresh + new chat
- At least portfolio + one other agent route
- API-only: accounts/holdings/imports if in scope

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